synchronous call

background or outbound

SecondBrain / JARVIS OS — System Architecture

Personal knowledge operating system: capture → memory → research layer → retrieval → generated answer

CLIENT · ELECTRON DESKTOP SHELL

React 18 + Vite UI
24 pages · 20 routes
Tailwind styling

Floating JARVIS orb
separate always-on-top
window · transparent

Electron main + preload
IPC: revealOrb · hideOrb
showMain · onWakeTrigger

Voice in the renderer
Web Speech API (STT)
speechSynthesis (TTS)

Context providers
BackendContext (JWT)
AssistantContext (socket)

server push: wake · speak

API LAYER · FASTAPI

REST /api/*
150 paths · 162 operations
20 route modules

WebSocket channels
/ws/live · /ws/voice · /ws/chat

Authentication
JWT (OAuth2 password)
per-request dependency

Background loops (asyncio)
mail sync · vault sync
embedding worker · OCR worker

CAPTURE & INGESTION

Sources and normalisation

- Screen capture (mss) → OCR (Tesseract)
- Mailbox over IMAP — read-only
- EXAMINE + BODY.PEEK, app password
- Google OAuth 2.0 (Gmail, Calendar)
- iCal / ICS feed parser
- Web page and social ingestion
- Voice: audio stream → VAD → STT
- Clipboard and file watchers
- — normalisation —
- Chunking · content hash · dedup key
- Writes Memory row + SearchIndex row

RESEARCH LAYER · THE PAPER'S CONTRIBUTION

Seven components (C1–C7)

- C1 Adaptive Memory Scoring — 8 auditable terms
- C2 Temporal Personal Knowledge Graph
- validity intervals · supersession · history
- C3 Contradiction-aware retrieval
- temporal · negation · value-change + damping
- C4 Memory Consolidation — 0.78 threshold
- provenance preserved on every merge
- C5 Knowledge-Gap Detection — exposure × depth
- C6 Retrieval-enforced Forgetting — 5 surfaces
- C7 PersonalBrain-Bench — evaluation instrument

PROACTIVE / HANDS-FREE

Always-on behaviours

- Wake word: openWakeWord
- 'hey_jarvis' at threshold 0.5
- Proactive voice policy:
- priority gate · quiet hours · dedupe
- Orb revealed over IPC, then spoken
- Turn-based mic loop: listen → think →
- speak → re-arm (onend watchdog)
- Mic released on shutdown so the probe
- server cannot crash the exit path

RETRIEVAL & GENERATION

writes memories

Four switchable pipelines
vanilla · hybrid · graph ·
adaptive (proposed)

Fusion and ranking
dense + lexical · 1/(1+i) prior
conflict / validity damping

Context assembly
retrieval-time dedupe
ordered, budgeted passages

LLM client (provider-agnostic)
local model over Ollama (11434)
or any OpenAI-compatible endpoint

PERSISTENCE · SIX STORES

SQLite — 49 tables
memory · capture · research
conversation · platform
single writer, WAL

Search index
full-text table joined
to every memory row

Vector store
Chroma embeddings
(optional; disabled in the
reported benchmark)

Memory vault (git)
cards/ · wiki/ · images/
knowledge_graph.json
auto-committed

Encrypted store
tokens and feeds (.enc)
key on disk, never
committed

HTTP (loopback)

git push / pull

EXTERNAL DEPENDENCIES

Google APIs
Gmail · Calendar (OAuth 2.0)

IMAP server
app-password mailbox

iCal / ICS feed
secret calendar address

GitHub
vault sync (deploy key)

Local LLM (Ollama)
e.g. qwen2.5:3b on RTX GPU